

ML Regression

Predicting a continuous target from 20 anonymized numeric features using cross-validated linear regression.

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Kaggle: Sofia ML Regression 2026 Summer · kaggle.com/t/9c82636ae0924188856c228fadfee148

0.541

Private R^2

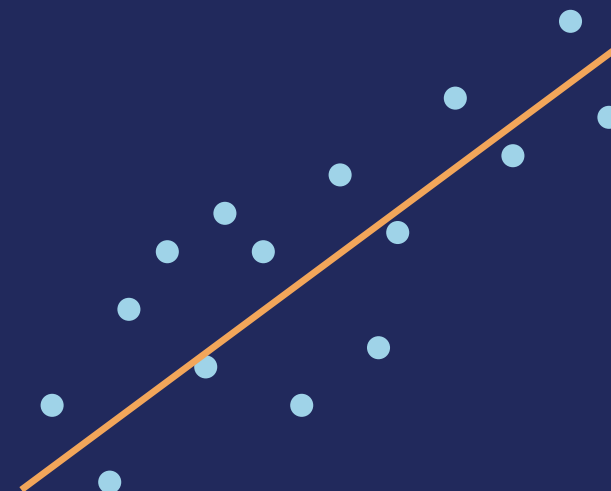
0.571

Public R^2

0.540

Posted Baseline
(Private R^2)

Private score edges past the posted baseline (0.541 vs 0.540)



Mean-imputed a linear-plus-noise dataset, then let simple Linear Regression outperform Ridge, degree-2 polynomial features, and a Random Forest in 5-fold CV — the winning move was restraint, not complexity.

A clean, anonymized, mildly noisy dataset



2,500 × 20

Rows and unlabeled numeric features (x0–x19) in both train and test sets.



~1% missing

Every feature has a scattering of missing values (15–31 rows) in train and test alike.



target $\approx N(1.8, 149)$

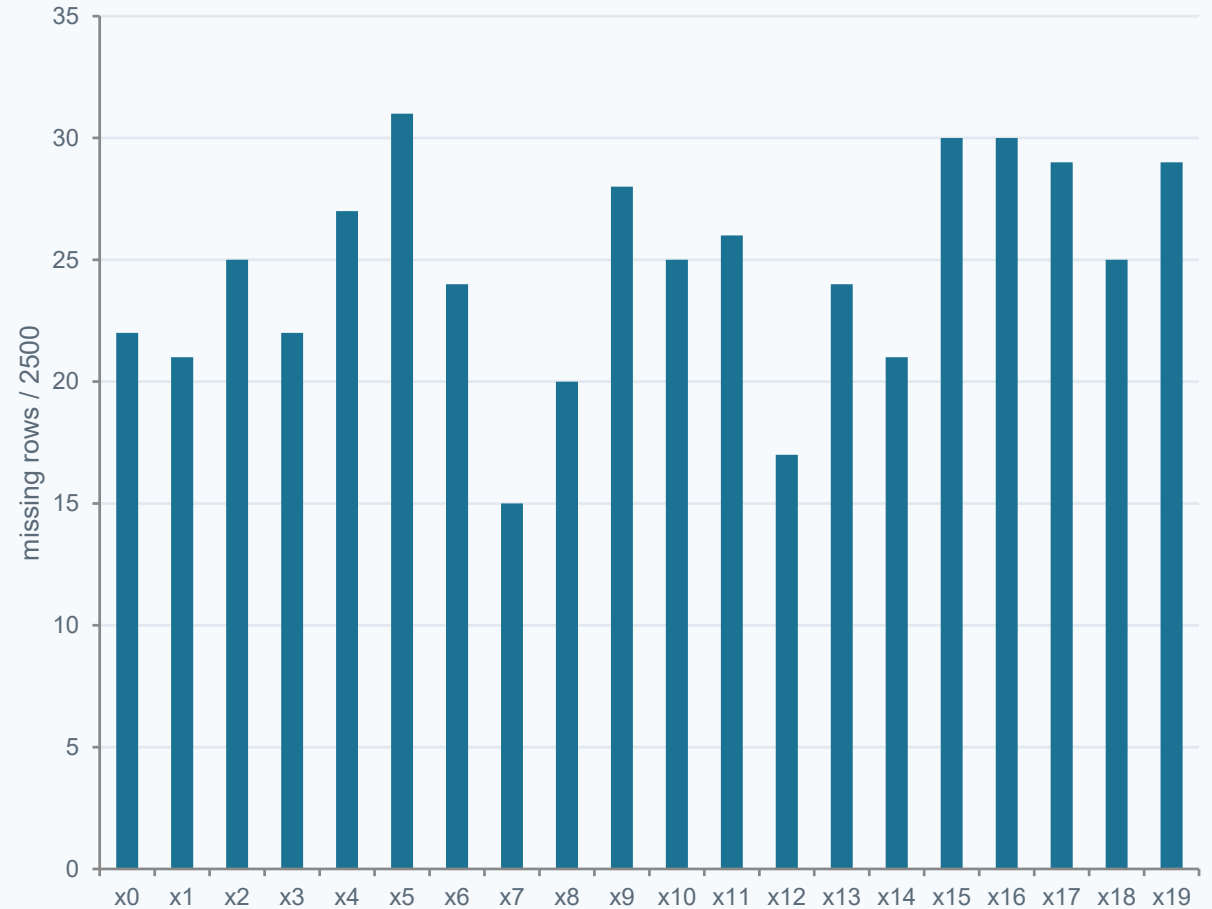
Continuous, roughly symmetric target ranging from -1215 to +968.



Near-linear signal

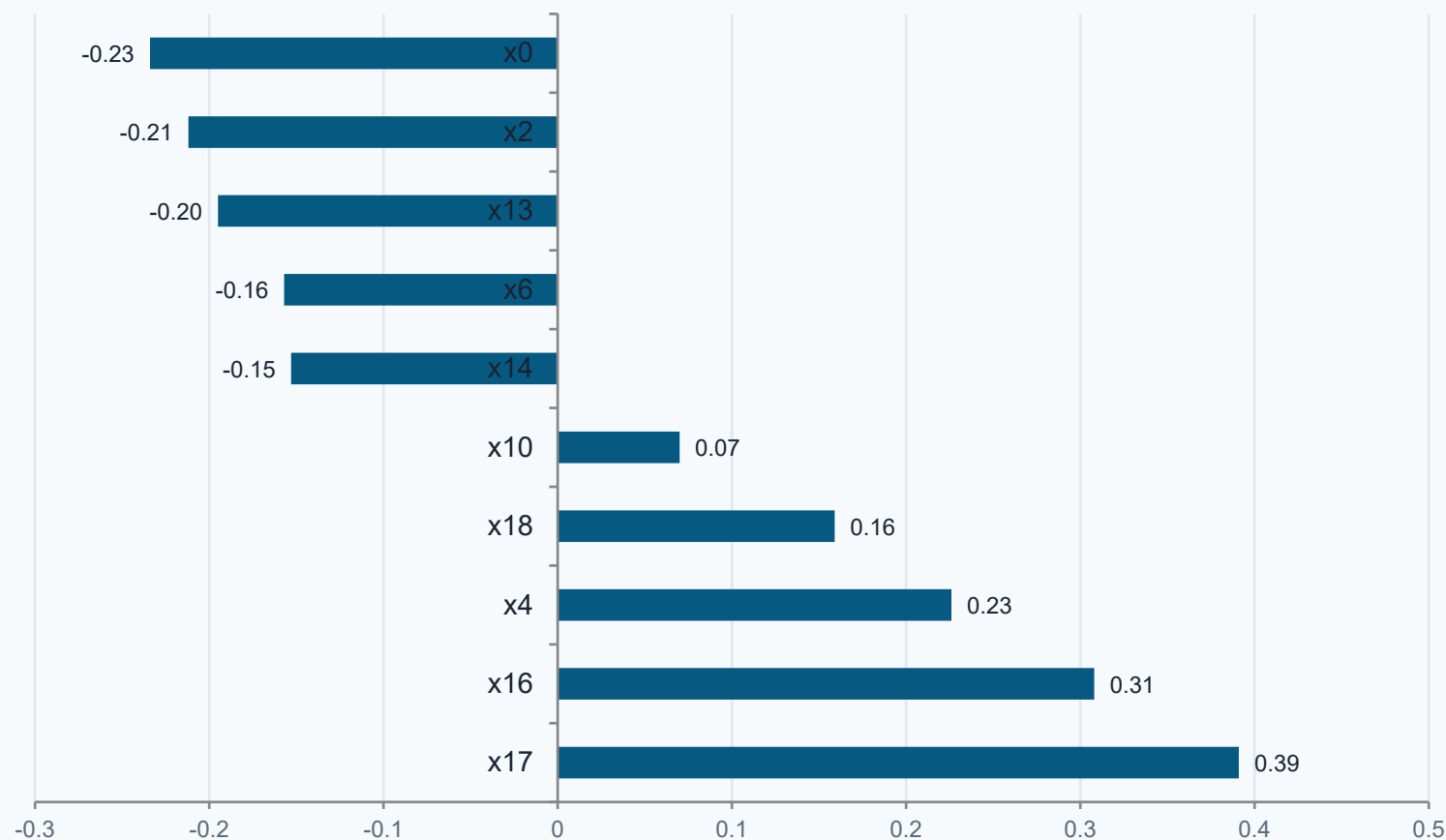
Correlation and modeling checks both point to target $\approx$ linear combination of features + noise.

Missing values per feature (train)



Which features move the target most?

Pearson correlation with target — top 5 positive and top 5 negative features



Reading the chart

x17 and x16 are the strongest single predictors ($|r| \approx 0.3\text{--}0.4$).

No feature exceeds $|r| \approx 0.4$ alone — the target depends on a **combination** of features, not one dominant column.

This motivated testing multivariate linear models before anything more complex.

Four approaches, evaluated with 5-fold CV

Linear Regression

Baseline: mean-imputed features, no regularization, no interactions.

0.540

CV R^2

Ridge (L2)

Tested $\alpha=1.0$ to see if shrinkage helps given noisy, correlated features.

0.540

CV R^2

Polynomial (deg. 2)

Added pairwise interactions + squared terms to probe for nonlinearity.

0.453

CV R^2

Random Forest

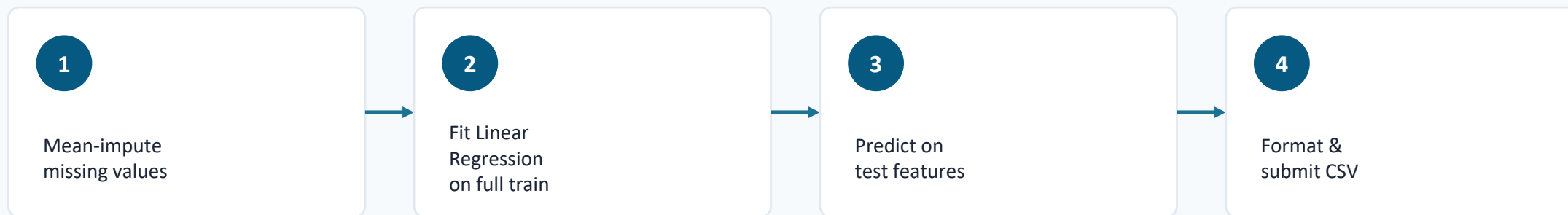
300 trees, to check whether a nonlinear model finds structure linear misses.

0.403

CV R^2

Motivation & takeaway: Polynomial features and Random Forest were tried specifically to test for nonlinearity or feature interactions — both underperformed the plain linear model, which is strong evidence the underlying process is linear plus noise. Regularization (Ridge) made no measurable difference, so the extra complexity wasn't needed either.

Pipeline, submission, and score



Results



Private score (0.541) edges past the posted baseline (0.540) — modest, but confirms the model generalizes rather than overfitting to the public split.

Everything above is reproducible

- **Kaggle notebook (EDA + all four experiments)**
kaggle.com/code/lydiasofia/YG-MSCS2202
- **GitHub repository (code + this slide deck PDF)**
github.com/<your-username>/sofia-ml-regression-2026
- **Competition**
kaggle.com/t/9c82636ae0924188856c228fadfee148

Key takeaway: when the data-generating process is close to linear, the winning model is often the simplest one — the job was validating that belief, not fighting it.



Thank you!!